Debugger

Debugging is a processes where the logical errors present in the pgm which are called as bugs can be removed

Bug is a logical error which leads to unexpected output of the pgm.

gcc filename.c → compilation

gcc -g filename.c → For debugging

gdb a.out

1. Breakpoint:

   break b linenumber

   b (f)

   b funname

2. run:

   r → for running your pgm

3. next:

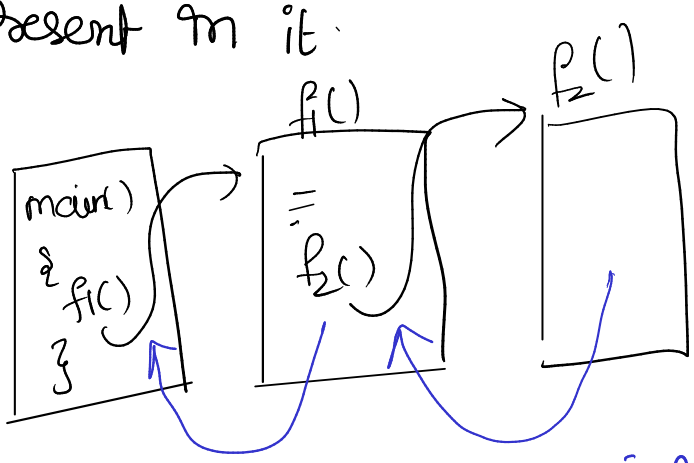
   n → for executing next line

4. print

   p → for printing what value the variable is holding by that line

5. backtrace:

bt to know where the fun' has to return after completing the execution of code present in it.



backtrace gives information related to Stack frame.

Online debugger

1. Stepover
2. Steptobo → executing line by line code
3. Stepout

```
main()
{
  f1()
}
```

```
f1()
{
  =
  f2()→Stepover
}
```

Stepout

```
f2()
{
  =
  =
  =
  =
}
```

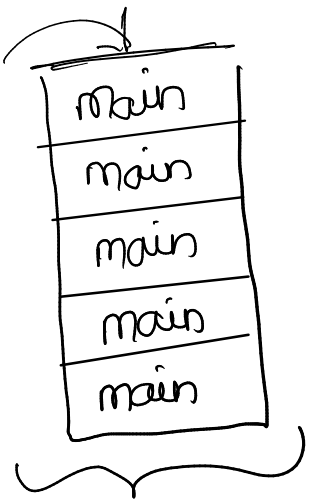
10th line

```
f2()
{
  =
  =
}
```

Recursion

A Function calling itself is called recursion

Ex:
```
main()
{
    main();
}
```

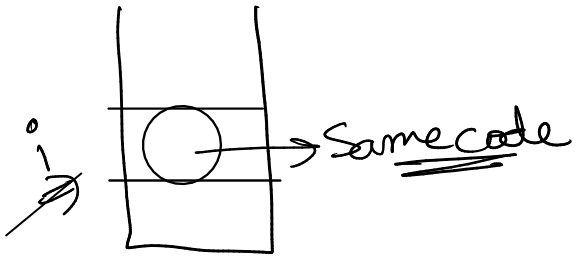


Segmentation fault

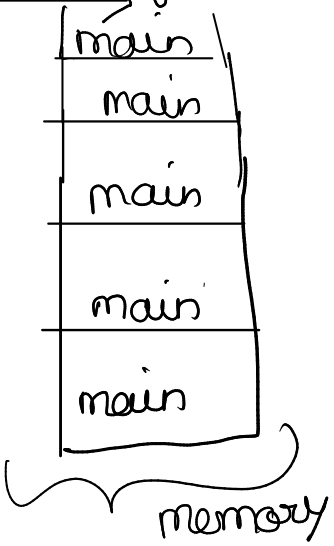
loop vs recursion

```
for(i=0; i<5; i++)
    printf("Hi");
```

```
int n;
main()
{
    printf("Hi");
    if(n<5)
    {
        main();
        n++;
    }
}
```



not possible using loops



used in link lists

Q) Wap to print the number in a range whose Setbit Count is even and the number should end with 4.